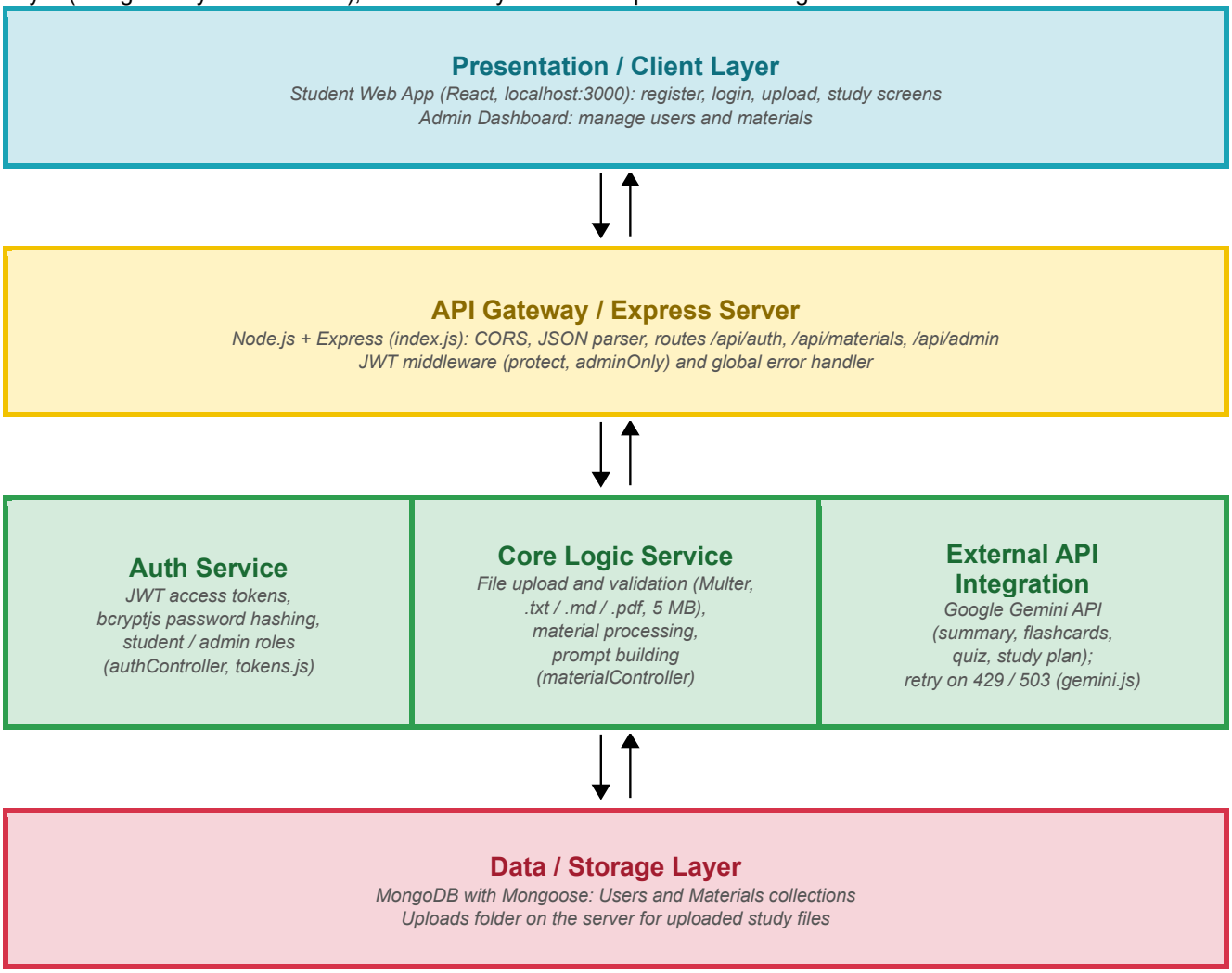
Solution Architecture

Date	30 September 2026
Team ID	SWTID-2026-2032
Project Name	Learnmate – AI StudyBuddy
Maximum Marks	5 Marks

Solution Architecture Diagram:

A high-level visual representation of the Learnmate architecture, mapping the Presentation Layer, Application Layer (API gateway and services), and Data Layer with the specific technologies used.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Web interface where students sign up, log in, upload notes and study the generated summary, flashcards, quiz and study plan; admins manage users and materials.	React.js (client at localhost:3000), HTML/CSS, JavaScript
API Gateway	Single entry point that receives client requests, applies CORS and JSON parsing, routes them to /api/auth, /api/materials and /api/admin, verifies JWTs and handles errors globally.	Node.js, Express.js, cors, express-async-errors, jsonwebtoken

Component Name	Description / Role in Architecture	Technologies Used
Auth Service	Registers users, verifies login, hashes passwords, issues JWT tokens and enforces student / admin roles.	JWT, bcryptjs, authController.js, tokens.js, auth.js middleware
Core Logic Service (Materials)	Validates and saves uploaded files (.txt, .md, .pdf, max 5 MB), builds AI prompts and stores the returned study content for each user.	Multer, materialController.js, upload.js
External API Integration	Sends prompts to Google Gemini and receives generated summary, flashcards, quiz and study plan; retries automatically on temporary 429 / 503 errors with exponential backoff.	Google Gemini API, @google/generative-ai, gemini.js
Admin Service	Admin-only endpoints to view and manage users and uploaded materials.	adminController.js, admin.js routes, adminOnly middleware
Database	Stores users (name, email, hashed password, role) and materials (content, summary, flashcards, quiz, study plan) with timestamps.	MongoDB, Mongoose (User and Material models), db.js
File Storage	Keeps uploaded study files on the server under unique file names.	Server uploads folder (Multer disk storage)